

A 12V-to-1~1.8V Tri-Path Series-Capacitor Converter with Reduced Inductor Current and Full-Range Duty Cycle for Point-of-Loads Application

Haixin Cai, Bin Li*, Zhaohui Wu

School of Microelectronics, South China University of Technology, Guangzhou, 510641, China

* E-mail: phlibin@scut.edu.cn

Abstract—This paper presents a tri-path series-capacitor buck converter with reduced inductor current and full-range duty cycle. Aiming for high step-down conversion ratio, the proposed converter uses three flying capacitors to relieve the switch voltage stress, making the low-voltage power MOSFET available. Besides, during the inductor demagnetization period, the flying capacitors constitute the capacitive current paths directly to the output, reducing the average inductor current and hence decreasing the conduction loss. Moreover, thanks to the single inductor and two-state operation per cycle, not only is the capacitor voltage automatically balanced, but also the inherent full duty cycle operation is achieved. Validated in a 180-nm BCD technology, the proposed converter regulates a 1 V – 1.8 V output voltage from a 12 V input voltage, and it achieves a high peak efficiency of 95.6%, with a switching frequency around 1 MHz.

Keywords—Buck converter, flying capacitor, reduced inductor current, full-range duty cycle.

I. INTRODUCTION

As the flourishing development of automotive, communication, and data center industries, power supply systems with an input voltage rail of 12 V are widely used, while the supply voltage of point-of-loads (PoL) has been reduced to 1 V or even lower due to the aggressive CMOS technology scaling [1]. Therefore, the power converters with a high step-down conversion ratio are in great demand to bridge the large voltage gap.

The conventional half-bridge buck converter, presented in Fig. 1(a), is the most commonly used since its simple structure and diverse control methods [2]. However, it is not suitable for PoL applications. First, the extremely short ON-time of no more than 1/12 due to the high step-down conversion ratio makes the design of the controller and the gate drivers quite challenging. Second, the bulky power devices with 12 V breakdown voltage have to be used, which adversely causes significant power loss and long delay. Finally, resulting from the large voltage swing at the switching node, a large inductor is needed to reduce the inductor current ripple, at the cost of a large volume.

To solve these problems, a double step-down (DSD) converter [3], as shown in Fig. 1(b), has been investigated in recent years. It uses a flying capacitor to sustain half of V_{IN} , alleviating the voltage stress on power switches to $V_{IN}/2$ and extending the duty cycle D . However, dual inductors for a large load current of $> 3A$ would lead to a large inductor DCR loss, and limit the duty cycle range as $D < 0.5$. To break the duty cycle limit, extra power switches [4] or capacitor cross-connected (CCC) topology [5] can be employed for $D > 0.5$ operation to achieve fast load transient response. However, the full-range duty cycle in steady-state operation is still not allowed, which limits the output voltage range.

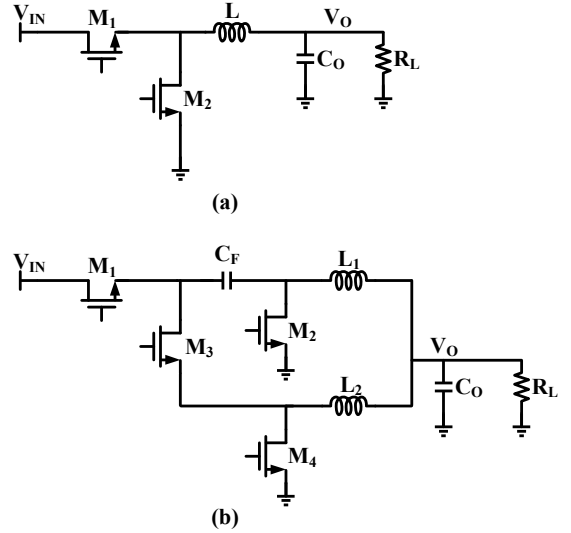


Fig. 1. (a) Conventional half-bridge buck converter, (b) double step-down (DSD) converter

To reduce the power loss and expand duty cycle range, this paper proposes a tri-path series-capacitor converter. The inherent full-range duty cycle is obtained due to the single inductor with two-state operation. Compared with the DSD converter, the flying capacitors not only reduce the switch voltage stress, but also provide additional current paths, subsequently reducing the inductor DCR loss.

The rest of this article is organized as follows. Section II discusses the operation principle of the proposed converter. Section III introduces the detailed circuit implementations. Then, simulation results are presented in Section IV. Finally, this paper is concluded in Section V.

II. PROPOSED TRI-PATH BUCK CONVERTER

Fig. 2 demonstrates the detailed power stage of the proposed tri-path series-capacitor converter which consists of 9 power transistors M_1 – M_9 , 3 flying capacitors C_{F0} – C_{F2} , and 1 inductor L . During the inductor magnetization period, the flying capacitors, similar to other hybrid buck converters, are used to reduce the input voltage stress. During the inductor demagnetization period, the flying capacitors constitute two extra current paths through the switch M_4 and M_9 respectively to the output, alleviating the inductor current stress. Moreover, with only one inductor, the proposed converter does not require phase interleaving. Therefore, a duty cycle beyond 0.5 is naturally supported.

A. Operation Principle

Fig. 3 illustrates the steady-state operation of the proposed converter, which has two operating states per cycle in continuous current mode. During phase Φ_1 , the inductor

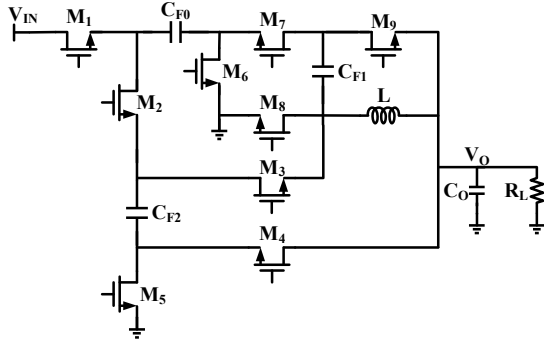


Fig. 2. Proposed tri-path series-capacitor converter

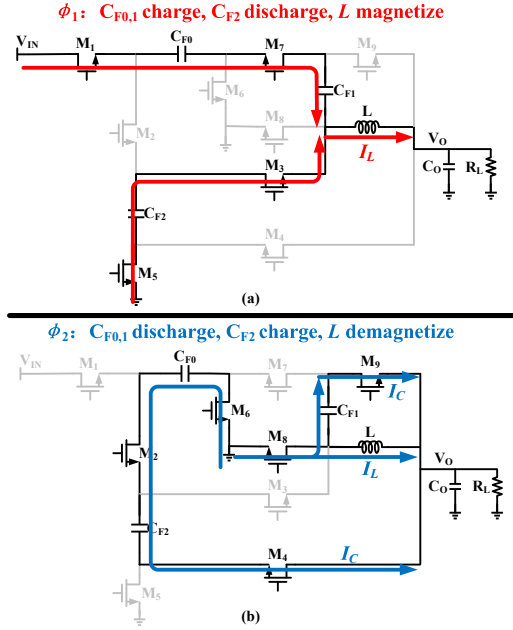


Fig. 3. Steady-state operation principle of the proposed converter

magnetization period, the switches M_1 , M_3 , M_5 , and M_7 turn on and switches M_2 , M_4 , M_6 , M_8 and M_9 turn off. In series with inductor L , the capacitors C_{F0} and C_{F1} are charged from V_{IN} , while C_{F2} discharges. During phase Φ_2 , the inductor demagnetization period, the switches M_2 , M_4 , M_6 , M_8 and M_9 turn on and switches M_1 , M_3 , M_5 , and M_7 turn off. In addition to the inductive-current I_L path, the capacitor C_{F1} , in parallel with the inductor L , provides an extra current path to the load. Moreover, the charge redistribution of C_{F0} and C_{F2} constitutes another capacitive-current I_C path.

Applying the voltage-second balance law to the inductor in the steady state, it can get

$$D(V_{IN} - V_{CF0} - V_{CF1} - V_{OUT}) + (1 - D)(-V_{OUT}) = 0 \quad (1)$$

with the duty cycle D defined as the ratio between the inductor magnetization period (state Φ_1) and the switching cycle T_s . Considering the charge redistribution between C_{F0} and C_{F2} during state Φ_2 , it can obtain

$$V_{CF0} = \frac{1}{2} V_{IN} \quad (2)$$

Similarly, since the flying capacitor C_{F1} redistributes charges with the output capacitor C_{OUT} at state Φ_2 , the voltage of the flying capacitor C_{F1} is

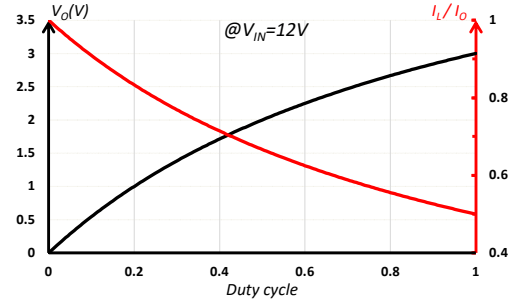


Fig. 4. V_O and I_L/I_O versus duty cycle.

$$V_{CF1} = V_{OUT} \quad (3)$$

The above analysis also means that the proposed buck converter has a self-balanced flying capacitor voltage, according to the charge distribution principle between parallel capacitors. Therefore, no additional flying capacitor voltage balancing loop is required.

By solving Eqs. (1), (2) and (3), the voltage conversion ratio (M) of the proposed converter can be obtained as

$$M = \frac{V_{OUT}}{V_{IN}} = \frac{D}{2(1 + D)} \quad (4)$$

Eq. (4) illustrates that the proposed converter can provide an output voltage less than $V_{IN}/4$, which is suitable for bridging the gap between 12 V input and < 3 V output with a significantly extended duty cycle. As Fig. 4 demonstrates, for a typical 12 V to 1 V or 1.8 V conversion, the duty cycle is extended to 0.2 and 0.43 respectively.

B. Inductor Current Analysis

By applying the charge balance law to three flying capacitors and the output capacitor in one periodic steady state, it leads to

$$D \cdot I_L + (1 - D) \cdot (I_L + I_{CF2,2} - I_{CF1,2}) = I_{OUT}$$

$$I_{CF0} = D \cdot I_{CF0,1} + (1 - D)I_{CF0,2} = 0$$

$$I_{CF1} = D \cdot I_{CF1,1} + (1 - D)I_{CF1,2} = 0 \quad (5)$$

$$I_{CF2} = D \cdot I_{CF2,1} + (1 - D)I_{CF2,2} = 0$$

where I_{CF0} (I_{CF1} , I_{CF2}), $I_{CF0,1}$ ($I_{CF1,1}$, $I_{CF2,1}$) and $I_{CF0,2}$ ($I_{CF1,2}$, $I_{CF2,2}$) are the average currents of C_{F0} (C_{F1} , C_{F2}) in one cycle, state Φ_1 , and state Φ_2 , respectively. Besides, by applying the Kirchhoff's Current Law at the switching node for each state, it can get

$$I_{CF0,1} = I_{CF1,1} = I_L + I_{CF2,1}$$

$$I_{CF2,2} = -I_{CF0,2} \quad (6)$$

By solving Eqs. (5) and (6), each flying capacitor current is obtained, and the average inductor current I_L can be written as

$$I_L = \frac{1}{1 + D} I_{OUT} \quad (7)$$

As shown in Fig. 4, with a VCR of 0.083 and 0.15, the proposed converter reduces the average inductor current by 16.6% and 30% respectively, compared with the classic half-bridge buck converter.

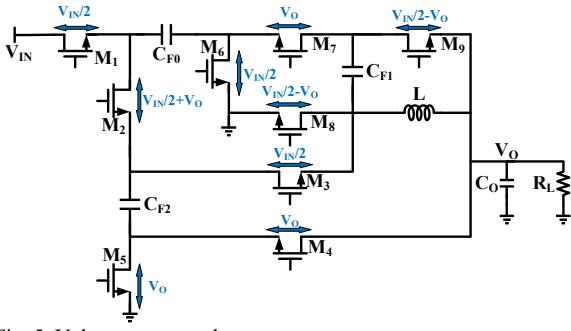


Fig. 5. Voltage stress on the power stage

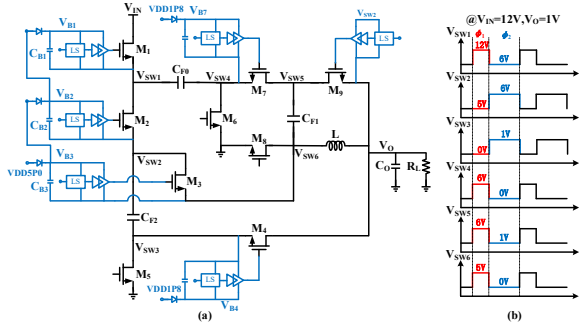


Fig. 6. (a) Power rails for drivers of each high-side switch (b) voltage swing of each switching node

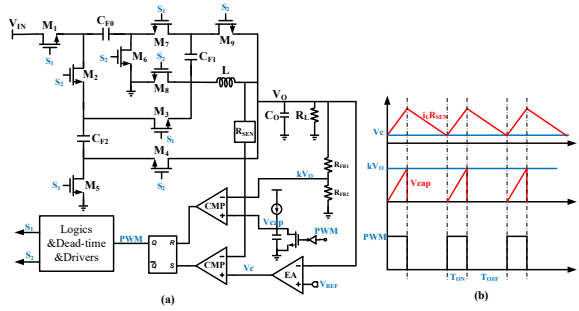


Fig. 7. (a) Overall control scheme of the proposed converter (b) key waveforms of controller in steady-state

III. CIRCUIT IMPLEMENTATION

The circuit implementation of the proposed tri-path converter is presented in this section, including the power stage, gate drivers, and the adaptive ON-time controller.

A. Power Stage

Fig. 5 indicates the detailed voltage stress of the power stage. Based on this analysis, appropriate transistor can be accordingly selected. To reduce power loss and area, all switches are implemented with N-type transistors. As shown in Fig. 5, only M_2 need bear the stress over $V_{IN}/2$, which is $V_{IN}/2 + V_O$. As a result of full-range duty cycle, the maximum V_O is 3 V for a 12 V input. Hence, M_2 is implemented with 9 V device. In addition, M_1 , M_3 and M_6 endure the stress of $V_{IN}/2$, while M_8 and M_9 are $V_{IN}/2 - V_O$, and the voltage stress across M_4 , M_5 , M_7 is V_O . For the typical 12V input 1-1.8 V output conversion, 6 V, 5 V and 1.8 V MOSFET switches can be used respectively. Compared with classic half-bridge and DSD converters, the proposed work successfully reduces the switch voltage stress, indicating an appreciable power efficiency.

B. Gate Drivers

Fig. 6 presents the voltage swing of each switching node and gives the detailed power rails for drivers of each high-side switch. As shown in Fig. 6, power rails for drivers of each high-side switch are implemented by bootstrap circuits. At Φ_2 , V_{SW6} is connected to ground and V_{B3} is charged to 5 V by VDD5P0. At Φ_1 , with V_{SW6} swinging to 5 V, V_{B3} is raised to 10 V, which can be used to charge C_{B2} directly because the source terminals of M_2 and M_3 are connected together at this time. Similarly, V_{B2} can also be used as the dynamic power rail for the M_1 driver because the source terminals of M_1 and M_2 are connected together at Φ_2 . Therefore, the stacked bootstrap design from VDD5P0 is employed, successfully reducing the design overhead. In addition, the power rails for M_4 and M_7 drivers are generated by two independent bootstrap circuits, where the bootstrap capacitors are charged by VDD1P8. Specially, M_9 can be powered by the switching node V_{SW2} directly considering that the voltage difference between V_{SW2} and V_O is enough to turn on M_9 , which further diminishes the design complexity.

C. Adaptive ON-time Controller

Fig. 7 depicts the overall control circuit implementations. Since the proposed converter does not require phase interleaving or capacitor voltage balance control, as commonly done by other series-capacitor converters, it can be easily applied in most advanced DC-DC controllers. Here, a valley current mode constant-on-time (COT) control is adopted for decent transient response.

Similar to current-mode PWM control, current-mode COT control introduces a current inner loop by sampling the ripple of the inductor current. The error amplifier (EA) compares the fraction of output voltage with the reference voltage V_{REF} to generate a control signal V_C , which is used to compare with the sampling voltage carrying the information of inductor current. When the inductor current drops below V_C , the PWM signal is flipped to a high level by the RS flip-flop, and the inductor begins to be magnetized to regulate output voltage. Meanwhile, the V_{CAP} starts to rise until it is above kV_O to adjust the ON-time.

Current-mode COT control requires Type-II compensation to stabilize the loop. Different from PWM control, COT control can attenuate the disturbance of the inductor current by cycle, eliminating the subharmonic oscillation. In addition, the introduction of a current inner loop allows the converter to operate normally with a low ESR output capacitor.

COT control also belongs to the PFM control mode, in which the switching frequency varies with the input and output voltage. By making the negative input of the comparator proportional to V_O and the positive input related to V_{IN} , an adaptive ON-time and a quasi-constant frequency of 1 MHz are achieved.

IV. SIMULATION RESULTS

The proposed converter is simulated in a 180nm BCD technology, where switches S_1 , S_3 and S_6 employ the 6 V LDMOS and switches S_8 and S_9 employ 5 V transistors. Since the voltage stress of switches S_4 , S_5 , and S_7 is V_O , 1.8 V transistors are employed in these switches. Moreover, S_2 is implemented with 9 V device since it sustains the stress of $V_{IN}/2 + V_O$. The inductor MAPM0420UA1R0M00-LF with an

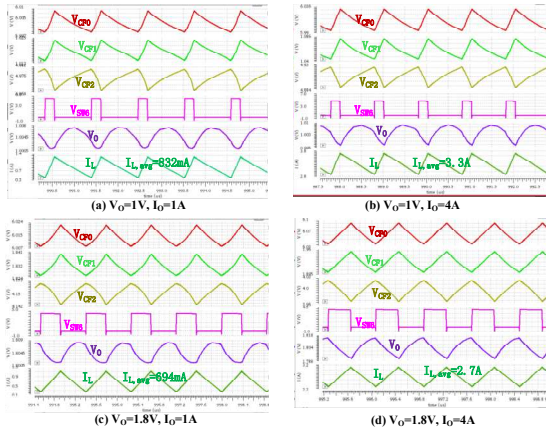


Fig.8. Steady-state waveforms at the different output voltages and currents

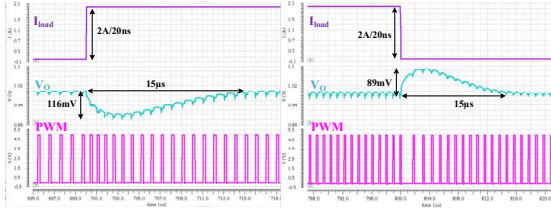


Fig.9. Load transient response waveforms

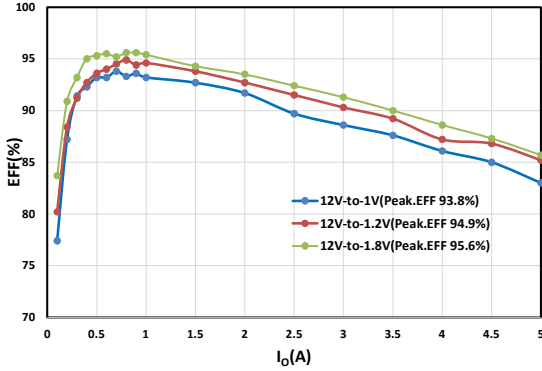


Fig.10. Efficiency versus load current for 12 V input and 1 V, 1.2 V, 1.8 V output

inductance of 1 μH and a DCR of 14 m Ω is used. Three flying capacitors C_{F0} , C_{F1} and C_{F2} are all set as 10 μF capacitance with ESR of 10 m Ω and the capacitance of C_{OUT} is set as 22 μF with ESR of 10 m Ω [7]. Moreover, the parasitic impedance of the package and PCB is set to 5 m Ω .

Fig. 8 plots the simulated steady-state waveforms at different output voltages and load currents. When $V_O = 1\text{ V}$, the average inductor current of the proposed converter under 1 A and 4 A loads is 832 mA and 3.3 A respectively, which is reduced by 16.7%. When $V_O = 1.8\text{ V}$, the average inductor current of the converter under 1 A and 4 A loads is 694 mA and 2.7 A respectively, which is reduced by 30%. Besides, the waveforms of V_{CF0} , V_{CF1} and V_{CF2} indicate that the voltages of the three flying capacitors are automatically balanced around $1/2V_{IN}$, V_O and $1/2V_{IN}-V_O$, respectively. Furthermore, the voltage swing of the switching node is successfully reduced, thus reducing the ripple and AC loss of the inductor.

Fig. 9 demonstrates the load transient response waveforms for a typical 12 V-to-1 V conversion, with a load current step between 1 A and 3 A and an edge time of 20 ns. For the up-transient, the proposed converter has an undershoot of 116 mV and a settling time of 15 μs . For the down transient, the

TABLE I. COMPARISON WITH STATE-OF-THE-ART CONVERTERS

	APEC' 16 [3]	ISSCC' 20 [2]	ISSCC' 22 [4]	ISSCC' 22 [5]	This work
Topology	DSD	Tri- State DSD	DSD	CCC	Tri-path SC
controller	AOT	Open Loop PWM	Voltage- mode PWM	Hysteretic	AOT
VCR	D/2	D/4	D/2	D/2	$D/2(1+D)$
V_{IN} (V)	12	12/24	12/24	12	12
V_O (V)	1.2	1	1	0.9-1.8	1-1.8
I_{O_MAX} (A)	10	3	4	4	5
F_{SW} (MHz)	2~5	0.1~1	1	2	0.9-1.2
L(μH)	2×0.22	2×0.56	2×1.8	2×0.74	1
C_F (μF)	2.2	2×1	4.7	2×2.2	3×10
C_O (μF)	350	10	10	10.6	22
Peak EFF @ V_O	87.7% @1.2V	91.2% @1.2V	88.3% @1V	86.8% @1.2V	95.6% @1.8V
Current Path	L	L	L	L	L+C
$D > 0.5$	NO	NO	NO*	NO*	YES

* $D > 0.5$ only in transient state

proposed converter has an overshoot of 89 mV and a settling time of 15 μs . As the transient waveform of PWM demonstrates, the AOT control can improve the transient response by reducing the OFF-time during the up transient and extending the OFF-time during the down transient.

Fig. 10 exhibits the power conversion efficiency versus the load current at 1 V, 1.2 V and 1.8 V outputs. The proposed converter achieves a high peak efficiency of 95.6% when the load current is 900 mA and V_O is 1.8 V. Besides, the converter has a load capacity of up to 5 A, with an efficiency of 85.7% at $I_O = 5\text{ A}$ and 83.7% at $I_O = 0.1\text{ A}$. The peak efficiency of the converter at V_O of 1 V and 1.2 V is 93.8% and 94.9%, respectively.

Table I is the comparison of the proposed converter with state-of-the-art hybrid converters. Benefiting from the reduced voltage stress and current stress by the proposed tri-path series-capacitor design, this work achieves a low VCR of $D/[2(1+D)]$ and favorable efficiency over a wide load range, even with only one inductor. In addition, without complicated phase shift or capacitor voltage balance techniques, control complexity is greatly reduced and inherent full duty cycle operation is obtained.

V. CONCLUSION

A novel tri-path series-capacitor buck converter is presented which not only reduces the voltage stress across the power switches and the current stress of the inductor, but also prolongs the duty cycle for the same voltage conversion ratio M , where $M = D/[2(1+D)]$. The proposed converter reduces the average current of the inductor by introducing two capacitive current paths, thus decreasing the conduction loss of the inductor. Besides, these flying capacitors are automatically balanced in the charge redistribution phase. Moreover, an adaptive ON-time control is employed for a quasi-constant frequency of 1 MHz. Based on 180-nm BCD technology, the simulation results show that the proposed

converter can regulate a 1 V – 1.8 V output voltage from a 12 V input voltage, with a 95.6% peak efficiency and an 85.7% 5 A-heavy-load efficiency.

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